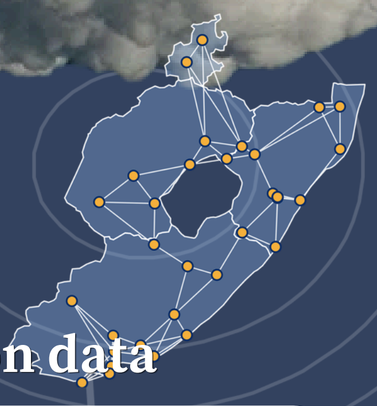
Weather Flow

Prediction using Spatial-Temporal Graph Neural Networks

Continuous-time, diffusion-based and lightweight models on South African station data



1 Overview

Weather at one station depends on its own recent past and on nearby stations. Spatial-temporal graph neural networks (STGNNs) capture both by treating stations as nodes in a graph. Prior STGNN studies successfully forecast South African weather up to 24 hours ahead. We test three newer STGNN families against the best models from that work, out to 48 hours.

2 Objectives

- Compare continuous-time, diffusion-based and lightweight STGNNs with the AGCRN and CLCRN baselines.
- Test how forecasts hold up as lead time grows to 48 hours and during extreme weather.

3 Methodology

Data

32 weather stations
10 years of hourly data
4 weather variables

South African Weather Service

Models

Baselines
AGCRN, CLCRN

Continuous-time
MTGODE, STGODE

Diffusion-based
DiffSTG, USTD

Lightweight
STLGRU, LiteSTGNN

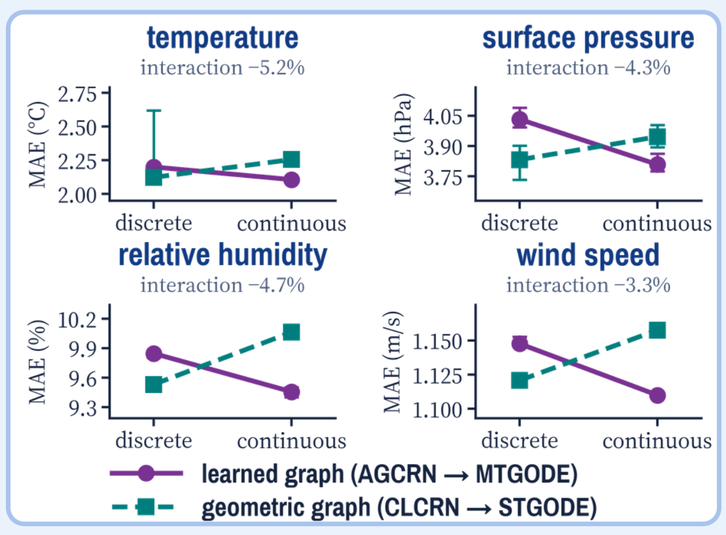
Evaluation

- Error to 48 hours ahead
- Extreme weather
- Uncertainty and cost

Held-out years, repeated runs

4 Results and analysis

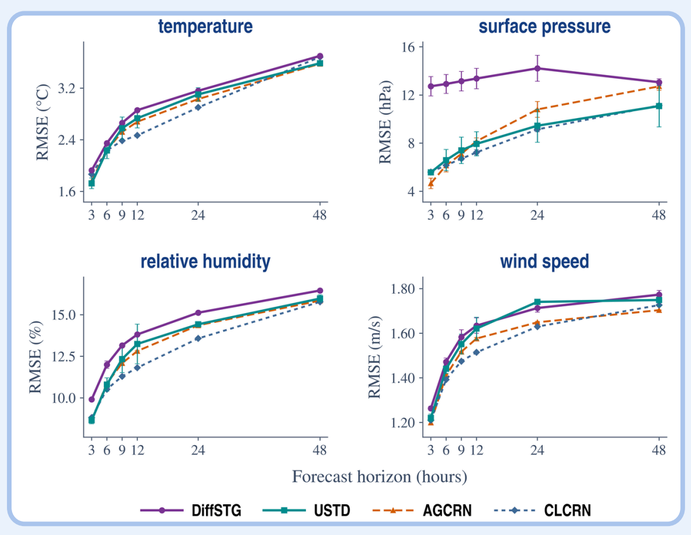
Continuous-time



The lines cross on every variable: continuous time helps with a learned graph and hurts with a geometric one.

Error is **3.5×** worse on extremes
MTGODE **12×** smaller than STGODE

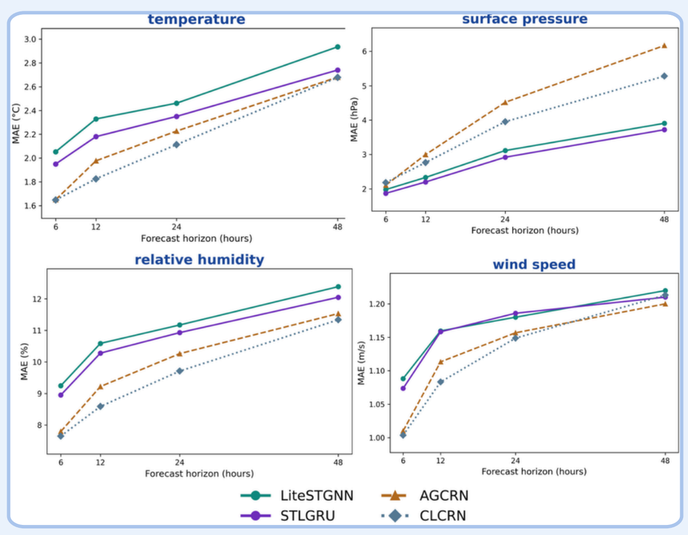
Diffusion-based



All models degrade as the horizon increases. CLCRN performs best on all variables, while DiffSTG performs the worst. USTD is competitive with AGCRN.

DiffSTG's average pressure RMSE is **1.5×** that of USTD's.

Lightweight



Adding more capacity to a lightweight model gives diminishing returns: STLGRU is **4.1x** larger than LiteSTGNN, but its accuracy gains are but improves MAE by only **0.7–5.7%.**

5 Conclusions

A new formulations from the baseline STGNNS (AGCRN & CLCRN) pay off only with the right graph and covariates.

1. Diffusion models did not consistently beat the baselines; continuous-time gains depended on the graph. Lightweight performance was competitive but variable-dependent.
2. Models converge by 48 hours, and strong average accuracy did not guarantee accurate forecasts of extremes.

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